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Estimating Anaerobic Threshold Using Brachial Artery Blood Flow Velocity During Incremental Cycling

A Pilot Study

Master's thesis in Physical Activity and Health

Supervisor: Ulrik Wisløff

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GRAPHICAL ABSTRACT

Study population



22 well-trained cyclists (17 men & 5 women)

Mean age: 25 ± 3 years

Mean VO_2peak : $61 \pm 7 \text{ mL}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$

Methods

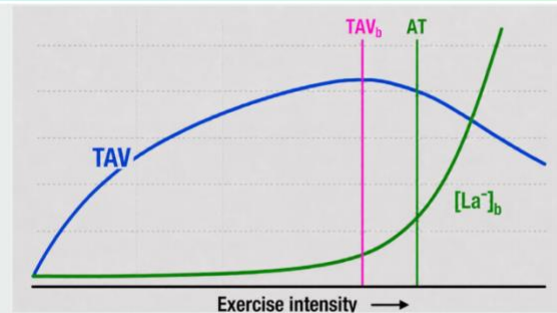
2x graded exercise tests



Continuous Doppler monitoring of brachial artery blood flow velocity (**TAV**)

Blood lactate sampling to determine **AT**

TAV_b versus AT



Validity vs AT
ICC 0.90 | $r = 0.95$

TAV_b occurred 12
W below AT

TAV_b reliability
ICC 0.73 | CV 8.4

Proposed mechanism: adrenergic activation



Sympathetic activation

Catecholamines $\uparrow$

Lactate $\uparrow$

TAV_b near AT

TAV_b was strongly related to AT, but occurred below and was not interchangeable with AT

ABSTRACT

Background: Anaerobic threshold (AT) is commonly assessed using blood lactate or ventilatory responses during incremental exercise testing, but these methods are invasive, episodic, or require specialized equipment. Doppler assessment of peripheral blood flow velocity may provide a non-invasive alternative. This pilot study evaluated the validity and test–retest reliability of the breakpoint of time-averaged brachial artery blood flow velocity (TAV_b) for estimating AT during incremental cycling.

Methods: Twenty-two trained cyclists completed two graded exercise tests. Blood lactate, heart rate, oxygen uptake, and brachial artery blood flow velocity were measured during exercise. TAV_b was identified automatically using a two-segment linear fit. Validity was assessed by comparing TAV_b with lactate-derived AT using intraclass correlation coefficients (ICC), Pearson correlations, and Bland–Altman analyses. Reliability between tests was assessed using ICC, coefficient of variation (CV), and Bland–Altman analyses.

Results: TAV_b occurred at significantly lower power output, heart rate, and oxygen uptake than AT. Agreement between TAV_b and AT was excellent for power output (ICC = 0.90), moderate for heart rate (ICC = 0.69), and good for oxygen uptake (ICC = 0.75). Pearson correlations between TAV_b and lactate-derived AT were strong for power output ($r = 0.95$), heart rate ($r = 0.77$), and oxygen uptake ($r = 0.80$). Reliability of TAV_b was moderate for power output (ICC = 0.73; CV = 8.4%) and poor for heart rate and oxygen uptake. AT showed good reliability for power output (ICC = 0.88; CV = 5.7%).

Conclusions: TAV_b showed moderate-to-good concurrent validity relative to lactate-derived AT. However, TAV_b occurred systematically below AT and showed only moderate reliability, which may reflect physiological variability and Doppler measurement limitations. Therefore, TAV_b should not be considered interchangeable with AT at present. Future studies should improve Doppler procedures and clarify the physiological basis and practical value of TAV_b.

DEFINITIONS

Anaerobic threshold – The highest exercise intensity during dynamic exercise with large muscle groups where production and clearance of lactic acid are approximately the same during a steady rate work condition.¹

Breakpoint of time-averaged velocity – The point during incremental exercise at which the time-averaged brachial artery blood flow velocity shows a breakpoint, identified automatically from the time-averaged velocity–time trace using a two-segment linear fit.

Graded exercise testing – An incremental exercise test in which exercise intensity is increased stepwise over time to assess physiological responses across increasing workloads. In the present study, this consisted of five-minute cycling stages with 25 W increments.

Maximal lactate steady state – The highest constant exercise intensity at which blood lactate concentration does not increase by more than 1 mmol·L⁻¹ between 10 and 30 min of exercise.²

Time-averaged velocity – The average blood flow velocity measured over time from the Doppler ultrasound signal of the brachial artery. In the study, this signal was measured continuously and smoothed using a 30-second trailing moving average.

Maximal oxygen uptake – The highest rate at which an individual can transport and utilize oxygen during dynamic exercise with large muscle groups, typically expressed in L·min⁻¹ or mL·kg⁻¹·min⁻¹.³

ABBREVIATIONS

AT – Anaerobic threshold

BMI – Body mass index

CV – Coefficient of variation

GXT – Graded exercise testing

HR – Heart rate

HR_{peak} – Peak heart rate

ICC – Intraclass correlation coefficient

[La⁻]_b – Blood lactate concentration

LoA – Limit of agreement

PO – Power output

PPO – Peak power output

SD – Standard deviation

SNS – Sympathetic nervous system

TAV – Time average velocity

TAV_b – Breakpoint of time-averaged velocity

VO₂ – Oxygen uptake

VO_{2max} – Maximal oxygen uptake

VO_{2peak} – Peak oxygen uptake

INTRODUCTION

Aerobic capacity is determined by three key physiological components: (i) maximal oxygen uptake (VO_2max), (ii) fractional utilization of VO_2max at anaerobic threshold (AT), (iii) and exercise economy.⁴ VO_2max is defined as the highest rate at which an individual can transport and utilize oxygen during dynamic exercise with large muscle groups,³ while exercise economy describes the oxygen cost of producing a given speed or power output. AT is defined as the highest exercise intensity during steady-state dynamic exercise with large muscle groups at which lactate production and clearance are approximately balanced.¹ Together, these determinants influence the speed or power output that can be sustained at AT, an important proxy for endurance performance in events such as marathon running and cycling time trials.⁵

Beyond its value as a marker of endurance performance, lactate is also relevant for tracking physiological changes during exercise. Blood lactate concentration ($[\text{La}^-]_b$) reflects the dynamic balance between lactate production and clearance, which fluctuates with changes in exercise intensity and provides insight into the metabolic demands of exercise.⁶ Therefore, monitoring $[\text{La}^-]_b$ responses during exercise can provide insight into the athlete's physiological state and may help guide individualized training intensity and recovery strategies, thereby supporting the optimization of athletic performance.⁷

AT is commonly assessed during incremental exercise testing, using either serial blood lactate measurements or ventilatory responses.⁸ Although widely used, these conventional approaches have important practical limitations. $[\text{La}^-]_b$ measurements require repeated blood sampling and are therefore invasive and episodic, whereas ventilatory thresholds depend on, continuous gas exchange analysis, expensive equipment, specialized personnel, and laboratory-based testing conditions. These limitations may restrict the routine use of conventional threshold testing by athletes, coaches, and teams, particularly when assessments need to be repeated frequently, applied to larger groups, or performed in field-based settings.⁹ This has led to growing interest in simple, non-invasive monitoring technologies that can track physiological responses related to AT during exercise in more practical settings.

Several non-invasive approaches have been explored for estimating AT, including physiological responses based on electromyography,¹⁰ heart rate variability,¹¹ and muscle oxygen saturation.¹² These approaches have shown promising results, particularly those based on muscle oxygen saturation, which has demonstrated strong agreements with the second ventilatory threshold.¹² Nevertheless, the measurement and interpretation of these responses have important limitations. Electromyography and near-infrared spectroscopy may be affected by adipose tissue thickness and muscle morphology, which can reduce signal quality and complicate interpretation.^{13,14} Heart rate variability, derived from beat-to-beat RR-interval fluctuations, provides a non-invasive marker of exercise intensity, but threshold detection appears less robust at higher exercise intensities and shows considerable inter-individual variability.¹⁵

In contrast, Doppler-based blood-flow assessment may provide an alternative approach, as ultrasound allows deeper tissue penetration and enables direct assessment of arterial flow velocity.¹⁶ Together with recent advances in compact Doppler ultrasound devices,¹⁷ this creates an opportunity to investigate Doppler-based haemodynamic monitoring as a potentially wearable and systemic approach to AT assessment.

The sympathetic nervous system (SNS) is a key regulator of cardiovascular responses during exercise.¹⁸ At the onset of large-muscle dynamic exercise (i.e., cycling), SNS activity and cardiac output increase rapidly, and blood flow is redistributed from central organs and inactive skeletal muscles toward the active skeletal muscles to meet their metabolic and oxygen demands while maintaining mean arterial pressure.¹⁹⁻²¹ In inactive skeletal muscles (i.e., the upper limbs during cycling), this redistribution is predominantly mediated by adrenergic vasoconstrictor mechanisms.²²

During incremental exercise, lactate accumulation increases progressively and becomes more pronounced around AT. Several studies have shown a clear association between plasma catecholamines and lactate accumulation.^{23,24} Accordingly, increasing exercise intensity may alter the balance between local vasodilatory influences and sympathetic vasoconstrictor restraint in the inactive arm, potentially producing a breakpoint in brachial artery blood flow

velocity. However, whether such a breakpoint exists, is reproducible, and corresponds to AT has not been established.

Therefore, this study aimed to evaluate the validity and test–retest reliability of the breakpoint in brachial artery blood flow velocity for estimating AT during incremental cycling. We hypothesized that brachial artery blood flow velocity would show a measurable breakpoint corresponding close to the AT, thereby providing a potential non-invasive alternative to blood sampling for threshold assessment.

METHODS

Participants

Twenty-five cyclists (18 men, 7 women) aged 20-28 years were recruited from university sports clubs. Inclusion criteria required cycling training at least three times per week. All participants provided written informed consent prior to participation. This study was conducted in accordance with research guidelines from the Norwegian University of Science and Technology and the general guidelines for research ethics issued by the Norwegian National Research Ethics Committees.

Study design

This pilot study used a repeated-measures observational design to evaluate the validity and test-retest reliability of TAV_b during exercise testing. Each participant completed two laboratory visits separated by at least one rest day and no more than one month. Each visit consisted of a graded exercise test (GXT) followed by a maximal incremental exercise test. Body mass and height were measured prior to exercise testing. During exercise testing, $[La^-]_b$, respiratory variables, heart rate, and brachial artery blood flow velocity were measured.

Body mass and height

Body mass was measured to the nearest 0.1 kg using the InBody 770 (BioSpace, Seoul, South Korea), and height was measured to the nearest 0.1 cm using a stadiometer (Seca 220, Hamburg, Germany), with participants wearing light clothing and no shoes. Body mass index (BMI) was calculated as $kg \cdot m^{-2}$. Measurements were undertaken after at least two hours of fasting and with an empty bladder.

Physiological testing

All exercise tests were performed on a cycle ergometer (Lode B.V. Medical Technology, Groningen, The Netherlands). Each visit began with a 10-minute individualized warm-up at <11 on the Borg 6–20 scale.²⁵ The first $[La^-]_b$ sample was measured immediately after the warm-up. The GXT then proceeded directly from the same workload as the warm-up and consisted of five-minute stages with 25 W increments, allowing near steady-state $[La^-]_b$ responses at each workload.⁸ Capillary blood samples were obtained during the final 30 seconds of each stage without interrupting exercise and analyzed using a stationary lactate analyzer (Biosen C-line, EKF Industrial Electronics, Magdeburg, Germany). When $[La^-]_b$

exceeded warm-up + 2.1 mmol·L⁻¹, one additional stage was completed, after which the maximal incremental test started directly. The maximal incremental test began with a 50 W increase, followed by 25 W increments every minute until volitional exhaustion or when cadence fell below 70 rpm.

Gas exchange was measured during the final two minutes of each stage and continuously throughout the maximal test using a metabolic cart (Vyntus CPX, Jaeger, Hoechberg, Germany). Oxygen uptake was calculated from the final 30 seconds of each stage. As not all participants fulfilled the predefined criteria for VO₂max, defined as a plateau in oxygen uptake despite increasing workload together with a respiratory exchange ratio ≥1.05 at exhaustion,²⁶ and given the prolonged protocol duration,²⁷ peak oxygen uptake (VO₂peak) was used instead. VO₂peak was calculated as the mean of the three highest consecutive 10-second values obtained during the maximal test. Heart rate was recorded continuously using a chest strap monitor (Polar H10, Polar Electro Oy, Kempele, Finland). Stage values were obtained from the final seconds of each stage, and peak heart rate (HRpeak) was defined as the highest value recorded during the maximal incremental test.

Anaerobic threshold determination

AT was defined as the workload corresponding to a [La⁻]_b value 2.1 mmol·L⁻¹ above the post-warm-up value, derived from the individual third-order polynomial lactate–workload curve. This delta value has been suggested to correspond most closely with maximal lactate steady state.^{28,29} Heart rate and oxygen uptake at AT were determined by linear interpolation between the two consecutive workload stages. Due to the systematic difference between hemolyzed and non-hemolyzed [La⁻]_b measurements,³⁰ this AT corresponds approximately to warm-up [La⁻]_b + 1.5 mmol·L⁻¹ when expressed in non-hemolyzed blood.

Ultrasound doppler measurements and blood flow velocity breakpoint determination

Peripheral blood velocity was measured continuously in the brachial artery on the medial aspect of the left distal upper arm using a non-commercial RescueDoppler system (Cimon Medical AS, Trondheim, Norway) equipped with a 4.0 MHz rectangular transducer (**Figure 1**).¹⁷ Doppler recordings were processed in MATLAB (R2025b, MathWorks Inc., Natick, MA, USA) using EarlyBird (II 6.17a, Cimon Medical AS, Trondheim, Norway) to obtain the blood velocity signal from the spectral Doppler recording (**Figure 2**). The resulting signal was

smoothed using a 30-second trailing moving average. A linear breakpoint detection algorithm implemented in Python (3.14.2, Python Software Foundation, Wilmington, DE, USA) was applied to the time-averaged blood velocity–time trace from the start of the first GXT stage until the end of the maximal incremental test, excluding the warm-up. The algorithm automatically identified the time point corresponding to the breakpoint in time-averaged velocity (TAV_b) (**Figure 3**). The power output at TAV_b was calculated by linear interpolation between the power of the preceding stage and that of the stage in which the breakpoint occurred. Heart rate and oxygen uptake at TAV_b were determined by linear interpolation between the two consecutive workload stages.



Figure 1. Experimental set-up for brachial artery blood velocity assessment during the graded exercise test. The image shows the Doppler transducer secured on the medial aspect of the distal upper arm to enable continuous brachial artery blood velocity assessment during cycling exercise.

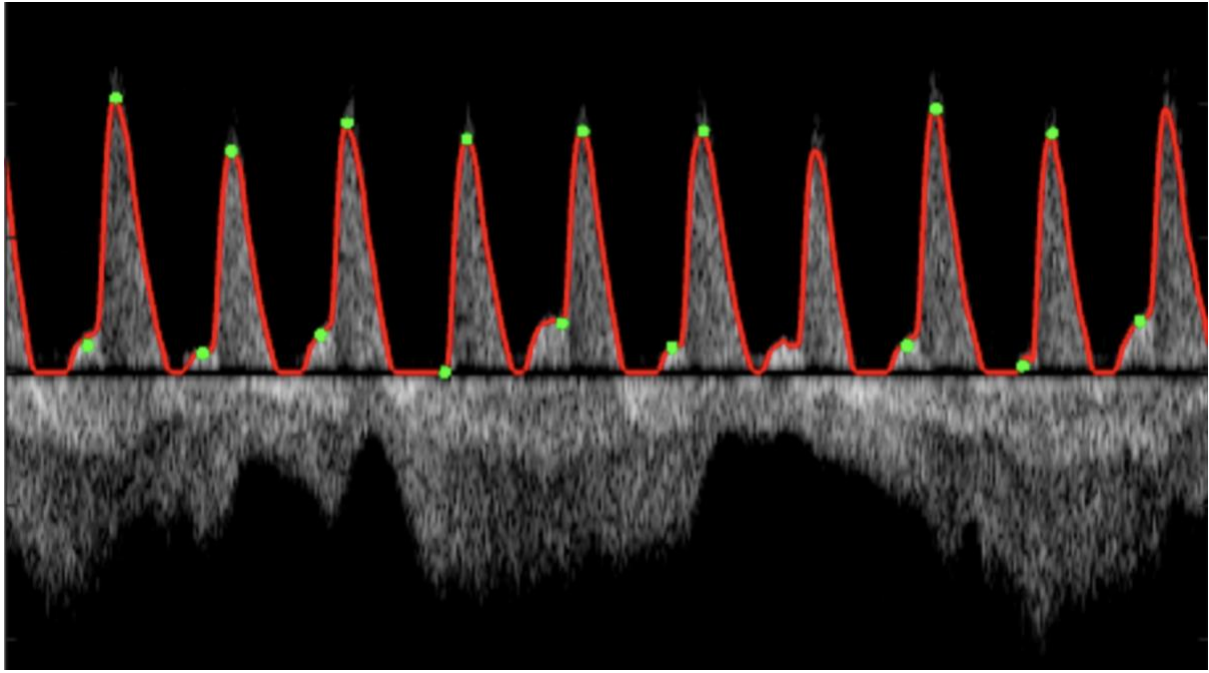


Figure 2. Representative spectral Doppler recording from the brachial artery during incremental cycling. The red tracing represents the derived blood velocity signal used for subsequent smoothing and breakpoint detection.

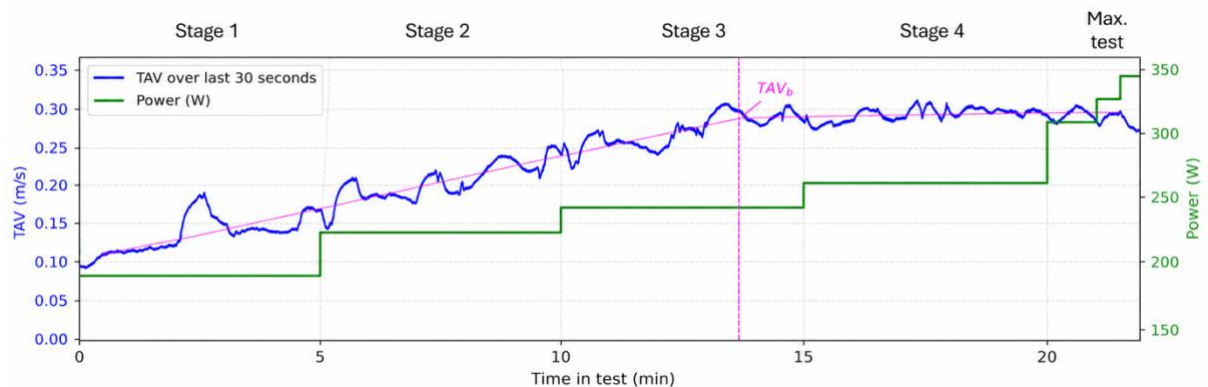


Figure 3: Linear breakpoint algorithm applied to TAV data in one test. The dark blue line shows TAV smoothed using a 30-second moving average, and the green stepped line shows power output (W). The breakpoint in the TAV data is extracted fully automatically using the breakpoint of a two-segment linear fit. The pink line represents the two-segment linear fit, and the dashed vertical pink line indicates the automatically detected breakpoint (TAV_b). The x-axis shows time in test (min). Stages 1–4 each lasted 5 min, followed by the maximal test starting at 20 min. Abbreviations: TAV, time-averaged velocity; TAV_b , TAV breakpoint.

Statistical analysis

Continuous variables are presented as mean $\pm$ standard deviation (SD) or median (range), as appropriate. Categorical variables are presented as counts. Normality was assessed using the Shapiro–Wilk test. For descriptive physiological variables and validity analyses, values from GXT1 and GXT2 were averaged for each participant. For the validity analyses, differences between TAV_b and AT were assessed using paired t-tests, and agreement was evaluated using intraclass correlation coefficient (ICC) estimates with 95% confidence intervals (CI), Pearson correlations, and Bland–Altman analyses. Intra-subject reliability between GXT1 and GXT2 was assessed separately for AT, TAV_b, and peak variables. Differences between tests were evaluated using paired t-tests. Agreement was assessed using Bland–Altman analyses and ICCs with 95% CIs. Spearman’s rank correlations were used to assess the associations between the number of days separating GXT1 and GXT2 and absolute test–retest differences in AT and TAV_b. The coefficient of variation (CV) was calculated to quantify within-subject variability. The size of ICC was evaluated as follows: ICC < 0.5 poor; ≥ 0.5 ICC < 0.75 moderate; ≥ 0.75 ICC < 0.9 good; and ICC ≥ 0.9 excellent.³¹ All statistical analyses were performed using Stata software (version 19; StataCorp, College Station, TX, USA). A two-sided p-value < 0.05 was considered statistically significant.

RESULTS

Four tests were excluded due to incorrect workload increments ($n = 2$) or missing TAV data ($n = 2$). Participants without two valid tests were excluded from further analyses. Consequently, 22 cyclists (17 men and 5 women) with two valid tests each were included in the primary analysis. The mean age of the final cohort was 25.1 ± 2.7 years, with a body mass of 76.1 ± 10.0 kg, height of 180.8 ± 8.8 cm, BMI of 23.2 ± 2.3 kg·m⁻², and VO₂peak of 61.4 ± 6.5 mL·kg⁻¹·min⁻¹. The third-order polynomial lactate–workload regressions showed consistently high model fit across included tests ($R^2 = 0.983 \pm 0.023$; range: 0.905–1.000). **Table 1** presents the physiological variables obtained at start GXT, TAV_b, AT, and peak exercise during GXT.

Table 1. Physiological variables at the start of GXT, TAV_b, anaerobic threshold, and peak exercise

	Start GXT	TAV _b	AT	Peak
PO (W)	177 ± 32	239 ± 36	251 ± 42	376 ± 49
PO (%PPO)	44.3 ± 4.8	63.9 ± 6.8	67.0 ± 7.6	100
HR (beats·min ⁻¹)	133 ± 8	157 ± 9	162 ± 10	189 ± 7
HR (%HRpeak)	70.1 ± 3.0	82.7 ± 4.3	85.1 ± 4.9	100
VO ₂ (mL·kg ⁻¹ ·min ⁻¹)	38.4 ± 6.0	43.8 ± 5.0	45.7 ± 5.4	61.4 ± 6.5
VO ₂ (%VO ₂ peak)	61.6 ± 5.1	72.3 ± 6.9	75.2 ± 5.0	100
[La ⁻] _b (mmol·L ⁻¹)	1.98 ± 0.83	—	4.08 ± 0.83	10.9 ± 2.5

Data are presented as mean ± standard deviation. Values represent the participant-level average of GXT1 and GXT2. Abbreviations: GXT, graded exercise test; TAV_b, time-averaged velocity breakpoint; AT, anaerobic threshold; PO, power output; PPO, peak power output; HR, heart rate; VO₂, oxygen uptake; [La⁻]_b, blood lactate concentration.

Validity

TAV_b occurred at significantly lower power output, heart rate, and oxygen uptake than AT ($p < 0.001$, $p = 0.017$, and $p < 0.001$, respectively). **Figure 4** shows scatter plots comparing TAV_b and AT for power output, heart rate, and oxygen uptake. ICC analysis indicated excellent agreement for power output (ICC = 0.90, 95% CI: 0.54–0.97), moderate agreement for heart rate (ICC = 0.69, 95% CI: 0.25–0.87), and good agreement for oxygen uptake (ICC = 0.75, 95% CI: 0.43–0.90). Pearson correlation analysis showed strong positive associations between TAV_b and AT for power output ($r = 0.95$, $p < 0.001$), heart rate ($r = 0.77$, $p < 0.001$), and oxygen uptake ($r = 0.80$, $p < 0.001$).

Figure 5 shows the Bland-Altman analyses for the comparison between TAV_b and AT for power output, heart rate, and oxygen uptake. For power output, TAV_b showed a mean bias of -12.1 W, with LoAs ranging from -39.5 to 15.4 W. For heart rate, the mean bias was -4.7 beats $\cdot$ min $^{-1}$, with LoAs ranging from -17.3 to 7.8 beats $\cdot$ min $^{-1}$. For oxygen uptake, the mean bias was -1.9 mL $\cdot$ kg $^{-1}\cdot$ min $^{-1}$, with LoAs ranging from -8.4 to 4.6 mL $\cdot$ kg $^{-1}\cdot$ min $^{-1}$.

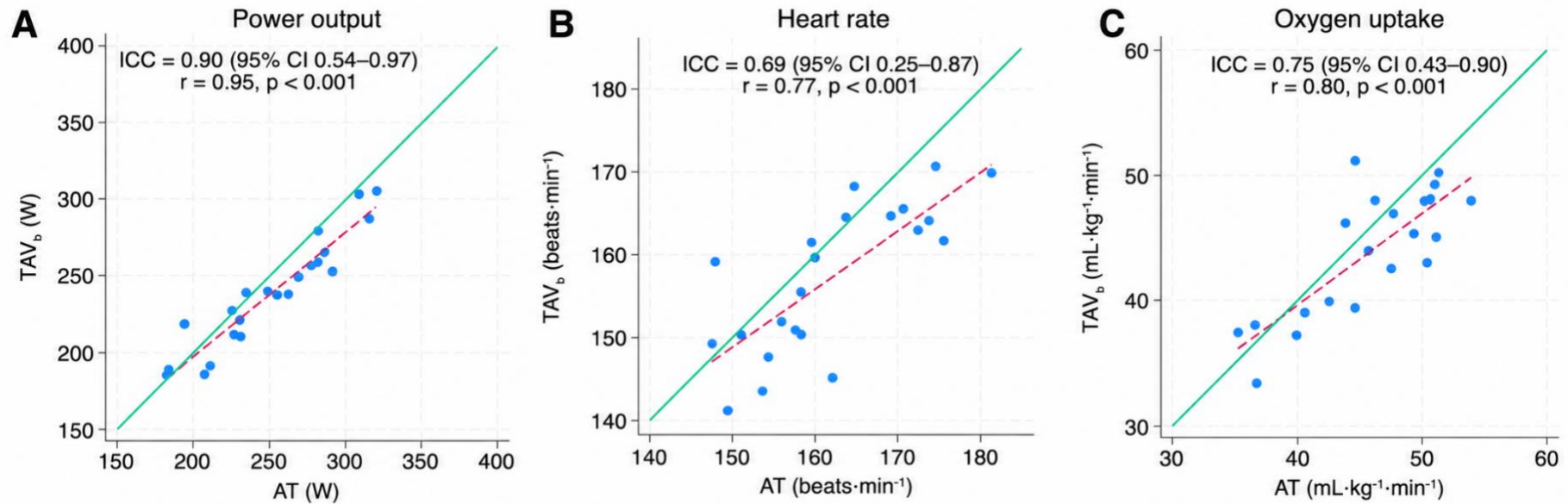


Figure 4. Scatter plots comparing TAV_b and AT for power output (A), heart rate (B), and oxygen uptake (C). Values represent the participant-level average of GXT1 and GXT2. The solid green line represents the line of identity ($y = x$), and the red dashed line represents the linear regression fit. The intraclass correlation coefficient with its 95% confidence interval and Pearson correlation coefficient are displayed within each panel. Abbreviations: TAV_b , time-averaged velocity breakpoint; AT, anaerobic threshold.

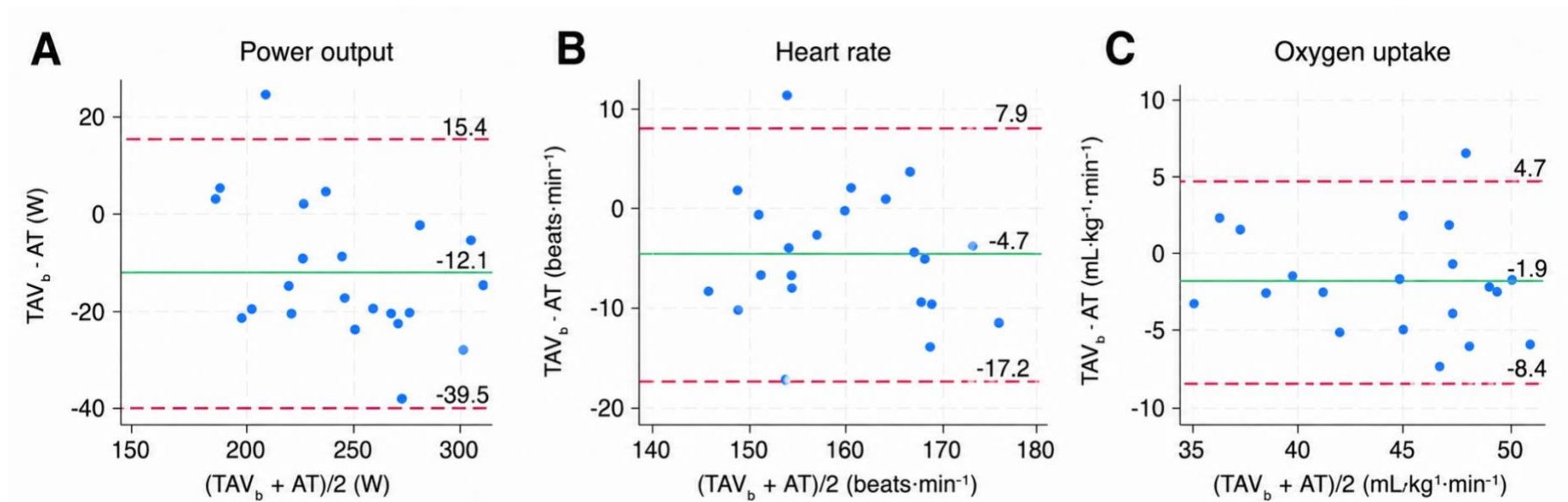


Figure 5: Bland–Altman plots illustrating the agreement between TAV_b and AT expressed as power output (A), heart rate (B), and oxygen uptake (C). Values represent the participant-level average of GXT1 and GXT2. The solid green line indicates the mean bias, and the red dashed lines indicate the upper and lower limits of agreement. The mean bias and limits of agreement are displayed within each panel. TAV_b , breakpoint of time-averaged velocity; AT, anaerobic threshold.

Reliability

Table 2 presents the test–retest reliability between GXT1 and GXT2 for AT, TAV_b, and peak variables, including ICCs, CVs, mean bias, and Bland-Altman analyses. For AT, power output demonstrated good test–retest reliability, with an ICC of 0.88 (95% CI 0.74–0.95), a CV of 5.7%, and a mean bias of –6.7 W. LoAs ranged from –46.4 to 33.0 W. Oxygen uptake at AT showed moderate reliability (ICC = 0.61, 95% CI 0.20–0.84; CV = 7.8%), whereas heart rate at AT showed poor reliability (ICC = 0.40, 95% CI 0.02–0.69; CV = 5.2%).

For TAV_b, power output demonstrated moderate reliability, with an ICC of 0.73 (95% CI 0.46–0.88), a CV of 8.4%, and a mean bias of 5.8 W. LoAs ranged from –50.0 to 61.5 W. Heart rate and oxygen uptake at TAV_b showed poor reliability, with ICC values of 0.34 (95% CI –0.11–0.67) and 0.38 (95% CI –0.18–0.74), respectively, and CVs of 5.8% and 9.8%.

Peak variables demonstrated moderate to good reliability overall, with ICC values ranging from 0.73 to 0.86 and CVs ranging from 1.6% to 5.3%. Peak power output showed a mean bias of –6.5 W, with LoAs ranging from –61.8 to 48.8 W.

Supplementary Figures S1 and S2 present the Bland-Altman analyses for test–retest agreement between GXT1 and GXT2 for AT and TAV_b, respectively, expressed as power output, heart rate, and oxygen uptake. The median time between GXT1 and GXT2 was 6.5 days (range: 1–31 days), and Spearman’s rank correlation analyses showed no consistent association between test interval and absolute test–retest differences in TAV_b or AT (**Supplementary Figures S3 and S4**). Individual responses across repeated GXTs are shown in **Supplementary Figure S5** and illustrate within-subject variation in TAV_b and AT.

Table 2: Reliability of TAV_b and AT within participants

	GXT1	GXT2	Paired samples t-test Mean difference (95% CI)	P-value	Bland-Altman LoAs	ICC ICC (95% CI)	CV %
AT							
PO (W)	254 ± 41	248 ± 44	-6.7 (-2.3, 15.6,)	0.137	-33.0, 46.4	0.88 (0.74-0.95)	5.7
HR (beats·min ⁻¹)	165 ± 11	159 ± 12	-6.4 (-11.7, -1.1)	0.020	-29.8, 17.0	0.40 (0.02-0.69)	5.2
VO ₂ (mL·kg·min ⁻¹)	46.8 ± 6.1	45.1 ± 5.4	-1.7 (-4.4, 1.0)	0.190	-11.6, 8.2	0.61 (0.20-0.84)	7.8
TAV_b							
PO (W)	236 ± 38	242 ± 40	5.8 (-6.8, 18.4)	0.351	-50.0, 61.5	0.73 (0.45-0.88)	8.4
HR (beats·min ⁻¹)	158 ± 9	157 ± 13	-1.2 (-7.0, 4.7)	0.684	-26.5, 24.2	0.34 (-0.10-0.67)	5.8
VO ₂ (mL·kg·min ⁻¹)	44.5 ± 5.0	44.4 ± 6.0	-0.1 (-3.5, 3.3)	0.960	-12.1, 12.0	0.36 (-0.07-0.68)	9.8
Peak							
PPO (W)	381 ± 53	376 ± 51	-6.5 (-19.7, 6.7)	0.316	-61.8, 48.8	0.86 (0.68–0.94)	5.3
HR _{peak} (beats·min ⁻¹)	190 ± 6	189 ± 7	-1.4 (-3.6, 0.7)	0.182	-9.6, 6.8	0.73 (0.40-0.89)	1.6
VO _{2peak} (mL·kg·min ⁻¹)	59.3 ± 6.1	60.6 ± 5.9	1.3 (-0.4, 3.0)	0.118	-5.2, 7.8	0.84 (0.60-0.94)	3.9

Mean differences and LoAs were calculated as GXT2 – GXT1. Abbreviations: TAV_b, time-averaged velocity breakpoint; AT, anaerobic threshold; LoAs, limits of agreement; GXT, graded exercise test; PO, power output; HR, heart rate; VO₂, oxygen uptake; PPO, peak power output; ICC, intraclass correlation coefficient; CV, coefficient of variation.

DISCUSSION

This pilot study aimed to evaluate the validity and test–retest reliability of TAV_b for estimating AT during incremental cycling. TAV_b showed strong agreement with AT, particularly for power output, and was strongly correlated with AT across power output, heart rate, and oxygen uptake. However, TAV_b occurred systematically below AT and showed only moderate reliability for power output, whereas AT showed good reproducibility. Reliability was lower for both thresholds when expressed as heart rate or oxygen uptake.

Validity

Our findings provide initial evidence that TAV_b has concurrent validity in relation to AT. The high agreement and strong Pearson correlations, particularly for power output, suggest that the Doppler-derived breakpoint is closely related to the lactate-derived AT. However, TAV_b occurred systematically below AT, indicating that the two methods should not be considered directly interchangeable at present. Rather, TAV_b may reflect a related physiological transition in brachial blood-flow velocity that occurs near, but not necessarily at, the lactate-derived AT.

Reliability

When expressed as power output, AT showed good test–retest reliability in the present study, with an ICC of 0.88 and a CV of 5.7%. Reliability estimates for lactate-derived thresholds vary both between studies and according to the threshold definition used. In a cohort similar to the present study, lactate thresholds expressed as workload showed CVs of 3.0–3.7% and ICCs of 0.85–0.89 in young trained male cyclists.³² Repeated incremental cycling tests in active but less-trained males showed ICCs of 0.65–0.90 and CVs of 3.8–12.1% across different lactate-derived threshold definitions.³³ Thus, the present AT reliability estimates fall within the previously reported range.

For TAV_b expressed as power output, test-retest reliability was moderate. This finding should not be interpreted solely as instability of the underlying physiological transition in brachial artery blood flow velocity, but may reflect a combination of Doppler-specific measurement variability and day-to-day physiological variation. During exercise, Doppler-derived velocity measurements were susceptible to probe movement, changes in insonation angle, and reduced alignment between the transducer and the artery, which may have altered the TAV signal and influenced breakpoint detection.

Reliability was lower when AT and TAV_b were expressed as heart rate or oxygen uptake. This may partly reflect day-to-day variability in heart rate and oxygen uptake at a given threshold workload, rather than variability in the threshold workload itself. Although no significant association was found between the number of days between GXTs and absolute test–retest differences, the non-standardized time interval may still have allowed day-to-day variation in training load, fatigue, nutrition, and recovery.^{33,34} These factors may have influenced heart rate, oxygen uptake, and lactate responses during exercise. In addition, the relatively high physiological responses at the start of the first GXT stage (i.e., 70% of peak heart rate and [La⁻]_b of 2.0 mmol·L⁻¹) suggest that lactate accumulation may already have begun in some individuals. Because AT was defined relative to post-warm-up [La⁻]_b, these higher lactate values may have shifted the individual AT criterion upward and influenced its estimation.

Sympathetic vascular restraint as potential mechanism underlying TAV_b

Our TAV_b may reflect the redistribution of blood flow during incremental exercise, driven by sympathetic vasoconstrictor restraint in inactive skeletal muscle. Previous cycling studies support the concept that blood flow in inactive limbs is restrained at higher exercise intensities. In ten healthy men performing sinusoidal leg cycling, brachial artery blood flow in the inactive upper limb was lower during higher-work-rate phases.³⁵ Additionally, in nine physically active men performing incremental cycling to exhaustion, subclavian blood flow remained relatively limited and arm vascular conductance was restrained, whereas blood flow to the active legs increased with exercise intensity.²⁰ At exercise onset, shear stress and endothelial nitric oxide release may contribute to the initial increase in TAV, reflecting a vasodilatory response in the brachial artery.^{36,37} As exercise intensity and active-muscle metabolic demand increase, greater blood-flow redistribution is required and sympathetic vasoconstrictor restraint becomes more pronounced. To meet these demands in active muscles, functional sympatholysis can overcome this vasoconstrictor influence, whereas in inactive muscles such vasodilatory influences are absent or far less pronounced.³⁸ Taken together, as exercise intensity increases during incremental exercise, rising SNS activity may contribute to a greater sympathetic vasoconstrictor restraint in inactive muscles, which could outweigh the initial vasodilatory response and is reflected in TAV_b.

Adrenergic activation as potential link between TAV_b and AT

A possible explanation for the observed association between TAV_b and lactate-derived AT is that both responses may be linked to increasing SNS activation and associated adrenergic stimulation during incremental exercise. Plasma catecholamines and blood lactate appear to increase in parallel as exercise intensity increases and have been reported to correlate closely.^{23,24} Because catecholamines stimulate muscle glycogenolysis and glycolysis,²³ ultimately increasing lactate production,³⁹ catecholamine release may contribute directly to the rise in [La⁻]_b during incremental exercise. β-adrenergic blockade studies support a role for catecholamines in lactate accumulation, although blockade does not completely prevent the rise in [La⁻]_b, indicating that other factors also contribute.^{40,41} For TAV_b, the vascular response may be more closely related to α-adrenergic vasoconstrictor restraint in the vascular bed of the inactive arm. The contribution of α-adrenergic mechanisms to sympathetic restraint during large-muscle dynamic exercise was demonstrated in a study of ten healthy men performing cycling exercise with and without combined handgrip exercise.²² In that study, cycling-induced vasoconstriction in the inactive forearm was attenuated by ~80% after local α-adrenergic receptor blockade. Taken together, these findings support the interpretation that TAV_b and lactate-derived AT may partly reflect a shared adrenergic response during incremental exercise, although neither response is likely to be explained by adrenergic activation alone.

Strength and limitations

A major strength of this pilot study is the novel definition and evaluation of TAV_b during incremental cycling. Because TAV_b was derived from continuous Doppler measurements of brachial artery blood-flow velocity, it may have potential relevance for future non-invasive AT assessment. However, several limitations should be considered. First, blood flow was inferred from TAV without assessment of arterial diameter and therefore may not reflect true volumetric flow, as velocity can be influenced by vessel diameter and flow patterns. This limitation is highlighted by previous work showing that brachial artery diameter can increase progressively by approximately 15% during prolonged cycling exercise.³⁶ Second, although a physiological mechanism involving SNS activation, catecholamines, and blood flow redistribution is proposed, no mechanistic intervention was used to directly test this pathway; therefore, the proposed mechanism remains speculative. Third, Doppler-derived TAV may

have been influenced by technological limitations during exercise, making it difficult to distinguish true physiological variation from measurement-related variability. Finally, the findings are limited to well-trained young cyclists and may not generalize to other populations, as both aging⁴² and exercise training⁴³ may modify sympathetic vasoconstrictor responsiveness in inactive skeletal muscle. Sex differences may also be relevant, as β -adrenergic mechanisms may play a more prominent role in women.^{22,44} However, it remains unclear whether these factors would alter the relationship between TAV_b and AT, as AT itself also shifts with age and endurance training.⁵

Future directions

Future studies should validate TAV_b in larger and more diverse populations and improve Doppler measurement procedures during exercise. Simultaneous assessment of brachial artery diameter and blood flow velocity is needed to determine whether TAV_b reflects changes in volumetric blood flow. Future work should also compare TAV_b with other established threshold markers, including maximal lactate steady state and ventilatory thresholds. Steady-state exercise studies may also help determine whether changes in blood lactate or metabolic status are reflected by corresponding changes in TAV.

Conclusions

The present findings support our hypothesis that brachial artery blood flow velocity shows a measurable breakpoint closely related to the lactate-derived AT. TAV_b showed good validity and may reflect a related physiological transition but occurred systematically below AT and should not be considered directly interchangeable. Adrenergic activation may contribute to the TAV_b response, although this was not directly tested. TAV_b showed only moderate reliability, which may partly reflect Doppler-specific technological limitations during exercise in addition to true physiological variation.

DISCLOSURES

During the preparation of this thesis, ChatGPT (GPT-5.5 Thinking; OpenAI, San Francisco, CA, USA) was used as a supportive tool for language editing, phrasing, and improving clarity and structure. The tool was not used to generate scientific interpretations or conclusions. All scientific content, interpretations, critical evaluations, and final decisions regarding the text were made by the author, who takes full responsibility for the final version of the thesis.

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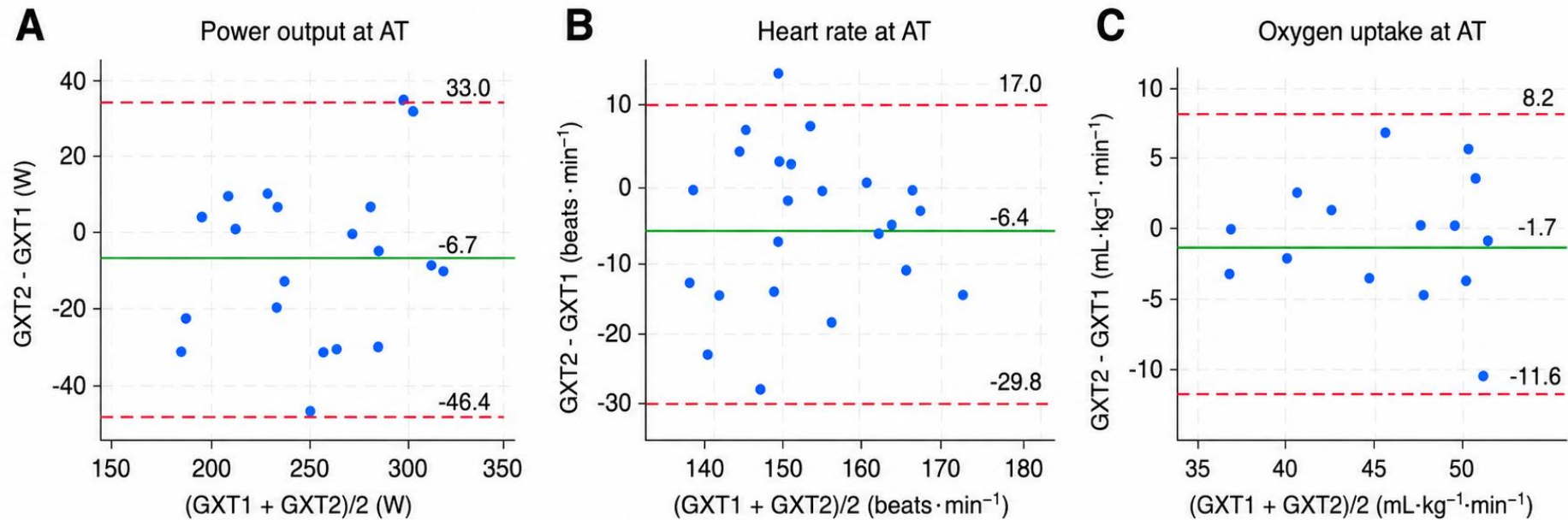
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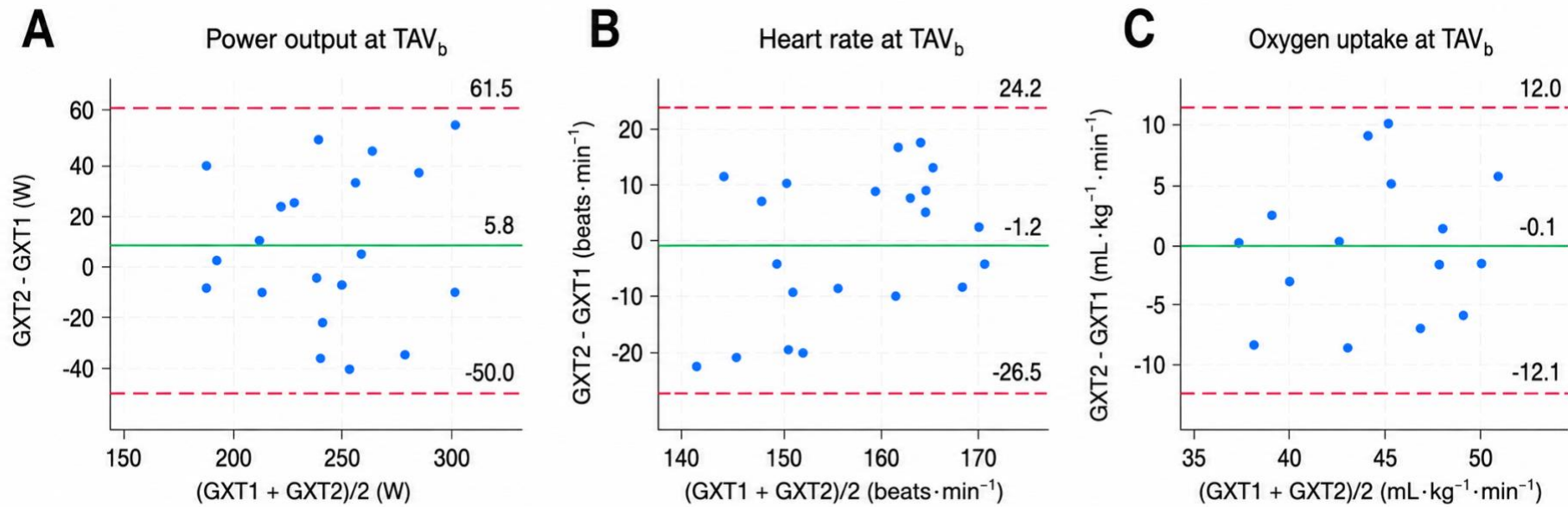
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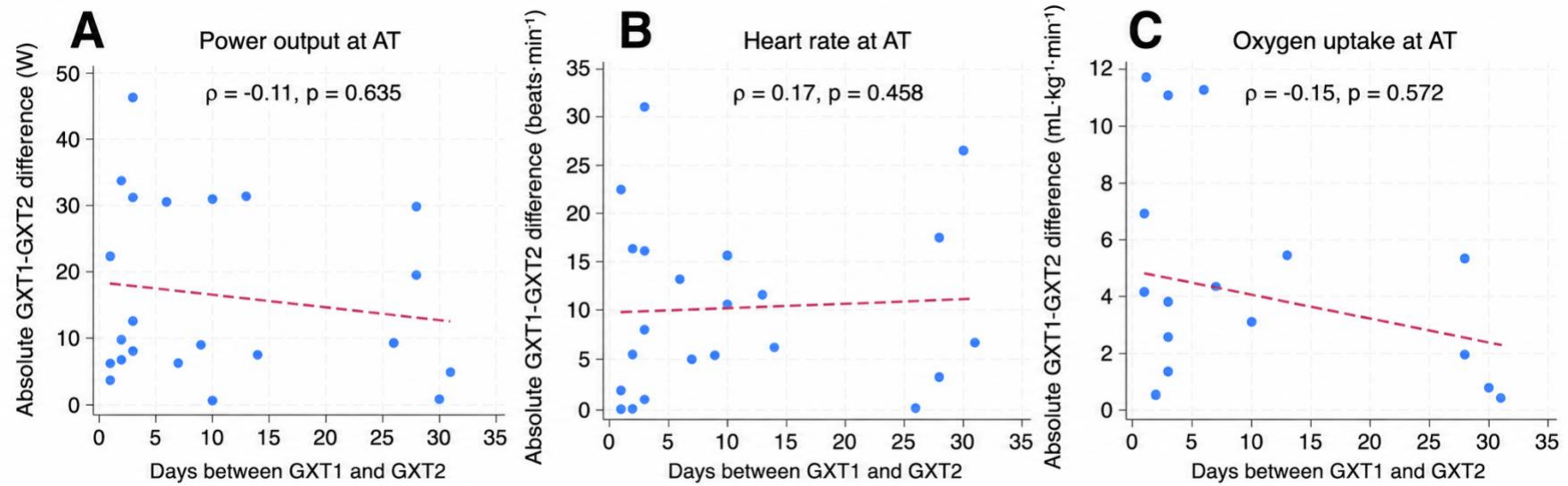
SUPPLEMENTAL MATERIALS



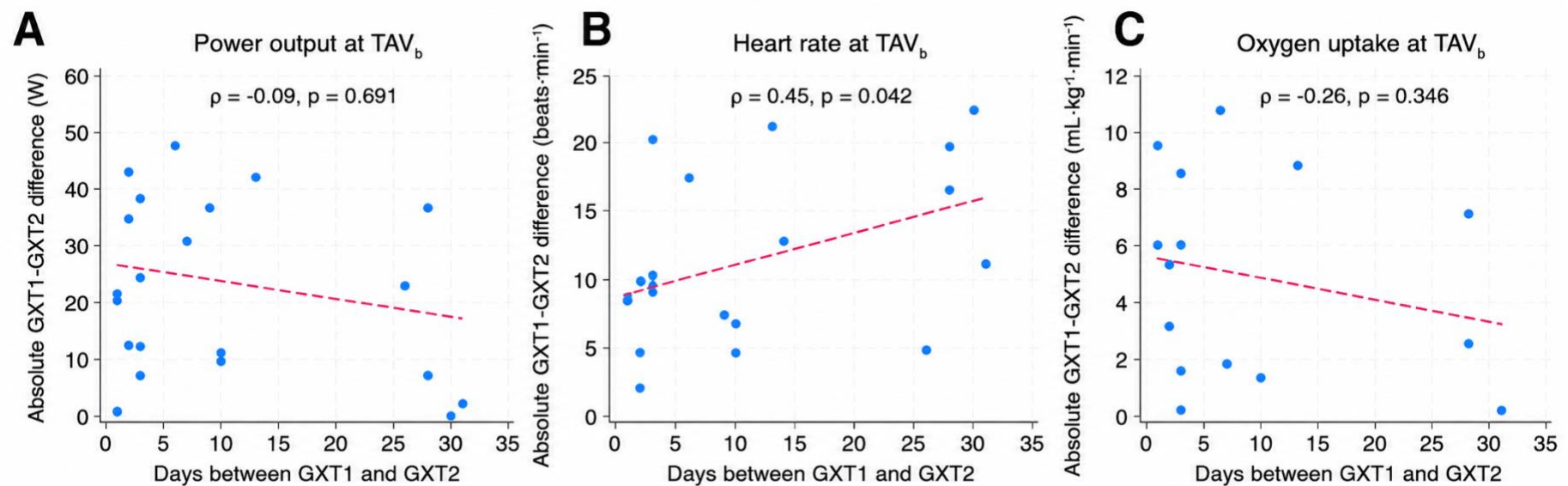
Supplemental Figure S1: Bland–Altman plots illustrating the intra-subject agreement between GXT1 and GXT2 for AT, expressed as power output (A), heart rate (B), and oxygen uptake (C). The solid green line indicates the mean bias, and the red dashed lines indicate the upper and lower limits of agreement. The mean bias and limits of agreement are displayed within each panel. AT, anaerobic threshold; GXT, graded exercise test.



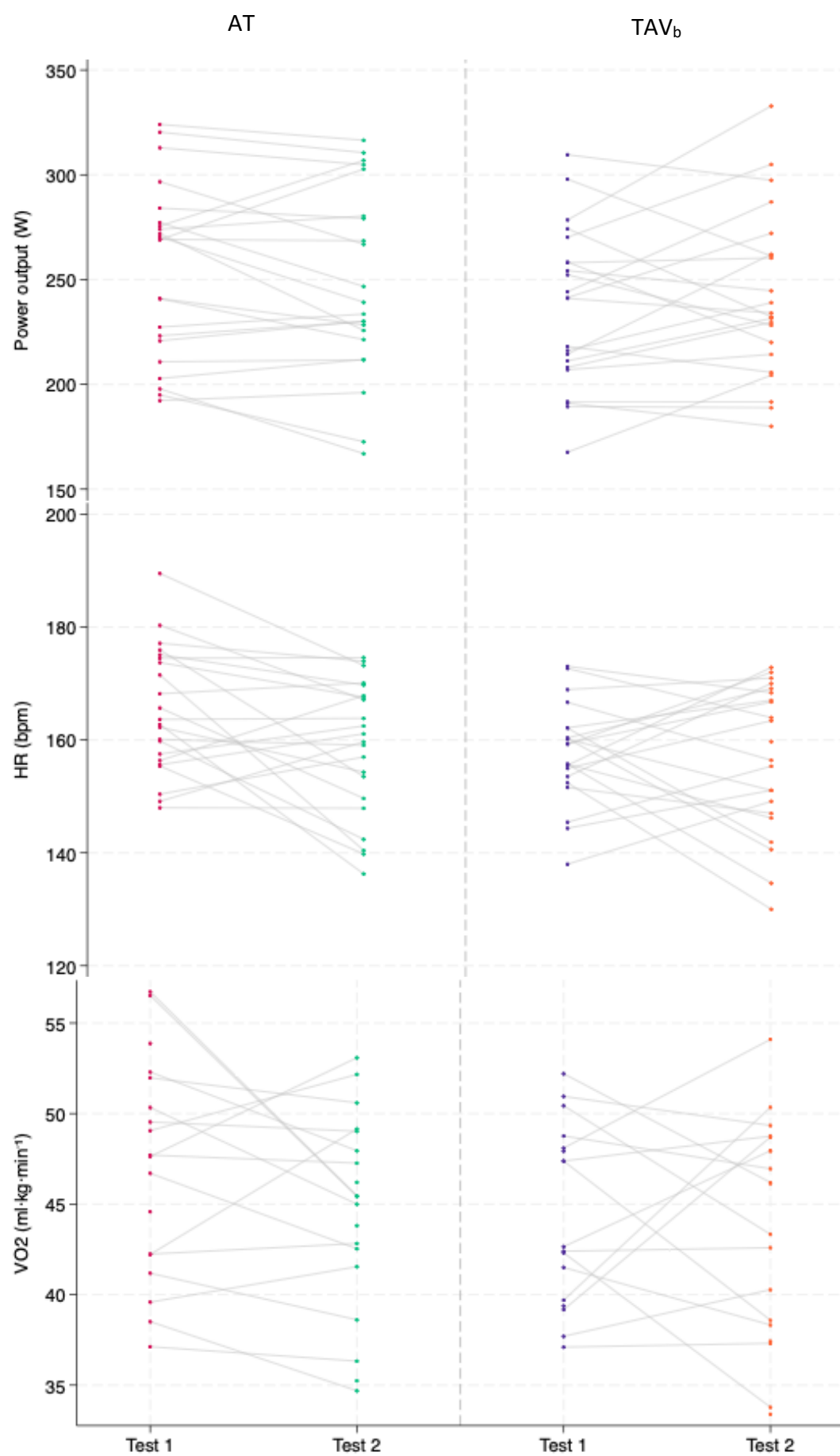
Supplemental Figure S2: Bland–Altman plots illustrating the intra-subject agreement between GXT1 and GXT2 for TAV_b, expressed as power output (A), heart rate (B), and oxygen uptake (C). The solid green line indicates the mean bias, and the red dashed lines indicate the upper and lower limits of agreement. The mean bias and limits of agreement are displayed within each panel. TAV_b, breakpoint of time-averaged velocity; GXT, graded exercise test.



Supplementary Figure S3. Scatter plots showing the association between the number of days between GXT1 and GXT2 and the absolute test-retest difference in AT for power output (A), heart rate (B), and oxygen uptake (C). Spearman's rank correlation coefficients (ρ) and p-values are displayed within each panel. AT, anaerobic threshold; GXT, graded exercise test.



Supplementary Figure S4. Scatter plots showing the association between the number of days between GXT1 and GXT2 and the absolute test-retest difference in TAV_b for power output (A), heart rate (B), and oxygen uptake (C). Spearman's rank correlation coefficients (ρ) and p-values are displayed within each panel. TAV_b, breakpoint of time-averaged velocity; GXT, graded exercise test.



Supplemental Figure S5: Individual response plots illustrating intra-subject changes between GXT1 and GXT2 for TAV_b (left panels) and AT (right panels), expressed as power output (top), heart rate (middle), and oxygen uptake (bottom). Each line represents one participant, connecting values obtained during GXT1 and GXT2. TAV_b , breakpoint of time-averaged velocity; AT, anaerobic threshold; GXT, graded exercise test.

